

Solfin- Report



Devisha Bhargava

Current Solution

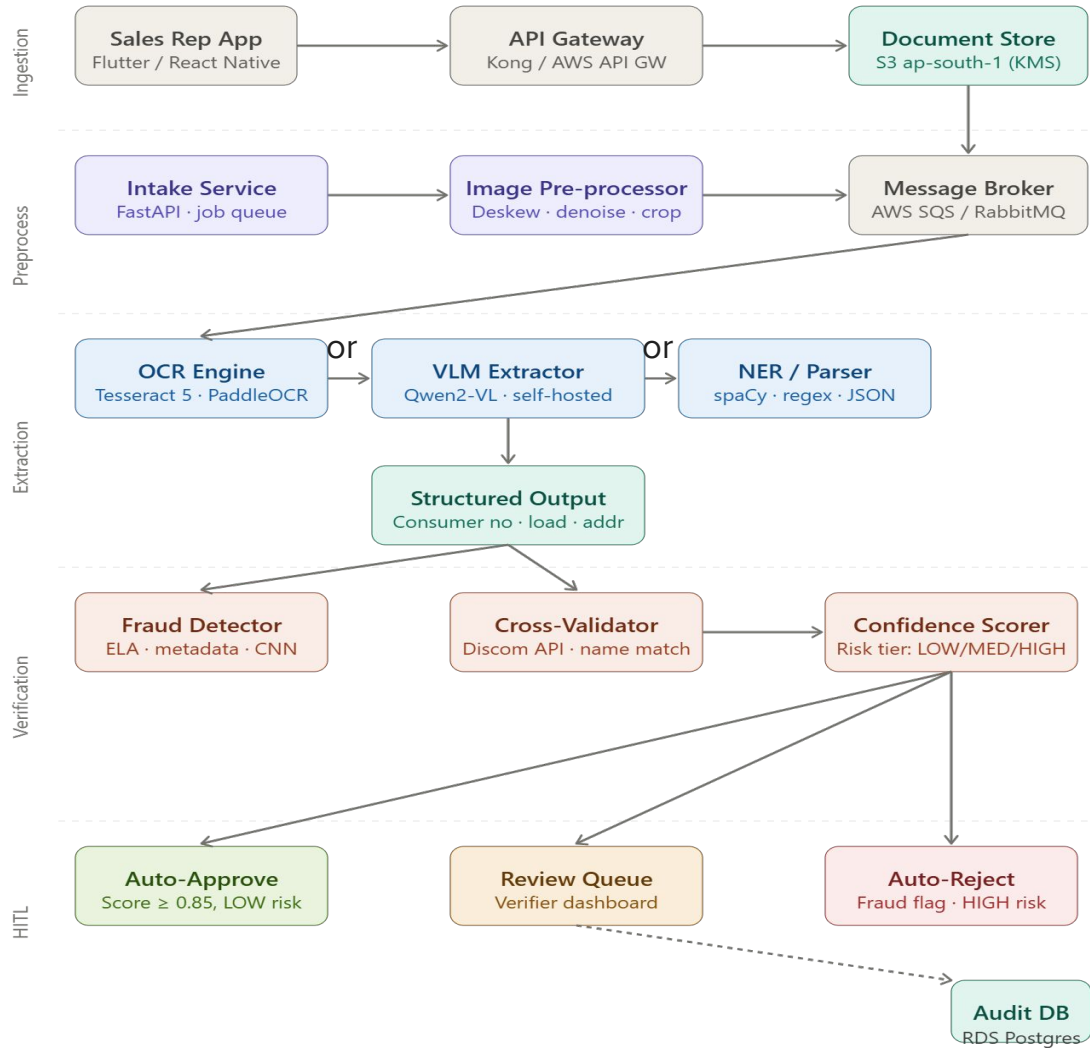
Solfin receives electricity bills from customers across India as part of its onboarding and verification workflow. These documents are submitted in multiple regional languages and formats, requiring the extraction of key data points for downstream decision-making.

The current extraction pipeline uses Google's Gemini API combined with prompt engineering to convert unstructured bill documents into structured data. Although functional, this architecture presents compliance and governance challenges:

- Customer documents containing sensitive personal information are shared with an external AI service.
- Processing may occur outside Indian data boundaries, creating data localization concerns.
- The approach may not align with organizational requirements for customer data privacy and regulatory compliance.
- Dependence on a third-party AI provider introduces operational and security risks.

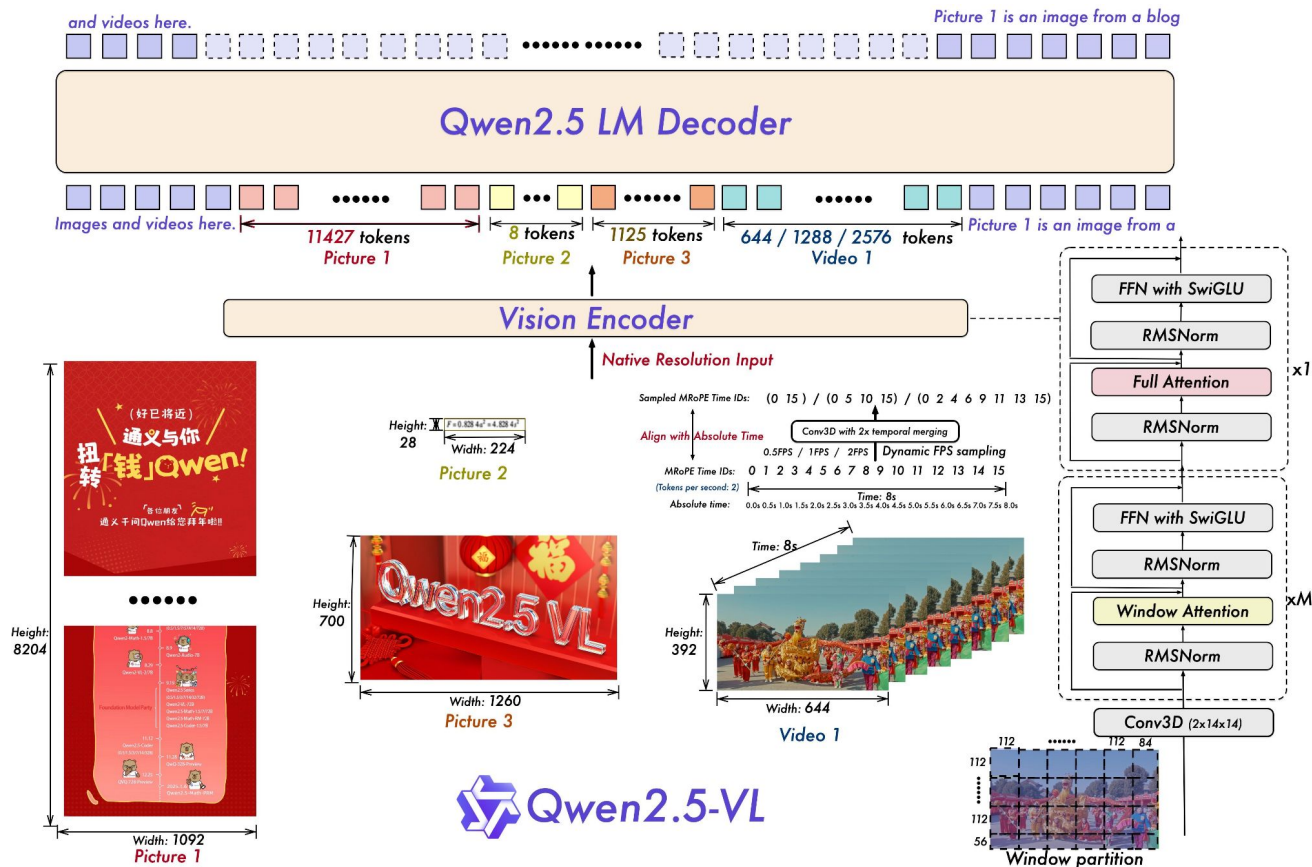
The objective is to design a solution that delivers high extraction accuracy while ensuring data sovereignty, regulatory compliance, and end-to-end protection of customer information.

Solution architecture as per instantial understanding



Week2

Understanding the working of Qwen 2.5 LLM Decoder



Qwen Performance for different document input

Agent benchmark

Benchmarks	Qwen2.5-VL-7B
ScreenSpot	84.7
ScreenSpot Pro	29.0
AITZ_EM	81.9
Android Control High_EM	60.1
Android Control Low_EM	93.7
AndroidWorld_SR	25.5
MobileMiniWob++_SR	91.4

Video Benchmarks

Benchmark	Qwen2-VL-7B	Qwen2.5-VL-7B
MVBench	67.0	69.6
PerceptionTest _{test}	66.9	70.5
Video-MME _{wo/w subs}	63.3/69.0	65.1/71.6
LVBench		45.3
LongVideoBench		54.7
MMBench-Video	1.44	1.79
TempCompass		71.7
MLVU		70.2
CharadesSTA/mIoU	43.6	

Week 3

Method 1: AWS Textract Queries API (Highly Recommended)

Instead of extracting all text and filtering it manually, AWS Textract has a built-in feature called **Queries** (part of the `analyze_document` API). You can pass natural language questions directly to Textract alongside your document, and it will return the exact answer along with its confidence score and bounding box.

Why it fits the use case:

- **High Accuracy:** It uses internal deep learning models to find the data regardless of where it's placed on the page.
- **Layout Agnostic:** If one utility company puts the "Pincode" at the top and another puts it at the bottom, Queries will still find it.

Method 2: AWS Textract Forms API (Key-Value Pairs)

Utility bills are essentially massive forms. Textract's FORMS feature automatically detects key-value pairs (e.g., it detects that `Bill Date :` is a key and `14/06/2026` is the value).

Pros & Cons:

- **Pros:** Completely automated; doesn't require predefined queries. It will return the bounding boxes for both the keys and the values.
- **Cons:** You will have to write code to map variations in keys (e.g., if one bill says `Consumer No .` and another says `Account Number`, you'll need a dictionary mapping to standardize them to your required `consumer_number`).

Method 3: The Bounding Box + LLM Method (Hybrid Approach)

If you still want to utilize standard OCR (`detect_document_text`) but need flawless accuracy, the industry standard is to extract the text blocks along with their layout/bounding boxes, and pass that structured payload to a Large Language Model (LLM).

Instead of just raw text, you pass the LLM a JSON structure representing the text words and their normalized coordinates, or you use a multimodal LLM (like Claude 3.5 Sonnet or GPT-4o) and pass it both the image *and* the raw text.

Which method will be best suitable

Method	Accuracy	Implementation Effort	Handling Layouts
Traditional Bounding Boxes (Hardcoded)	Low	High (Brittle)	Poor
AWS Textract Queries	Very High	Medium	Excellent
AWS Textract Forms (KV)	High	Medium	Good
OCR + LLM Parsing	Very High	Low	Excellent

What is AWS Textract Adapters?

AWS Textract Adapters are a feature of **Amazon Textract** that allows you to **customize Textract for a specific document type** without training and deploying a full machine learning model yourself.

Think of it this way:

- **Standard Textract** = generic OCR + form/table extraction that works reasonably well across many document types.
- **Textract Adapter** = a document-specific enhancement layer that learns the layout and fields of your documents and improves extraction accuracy.

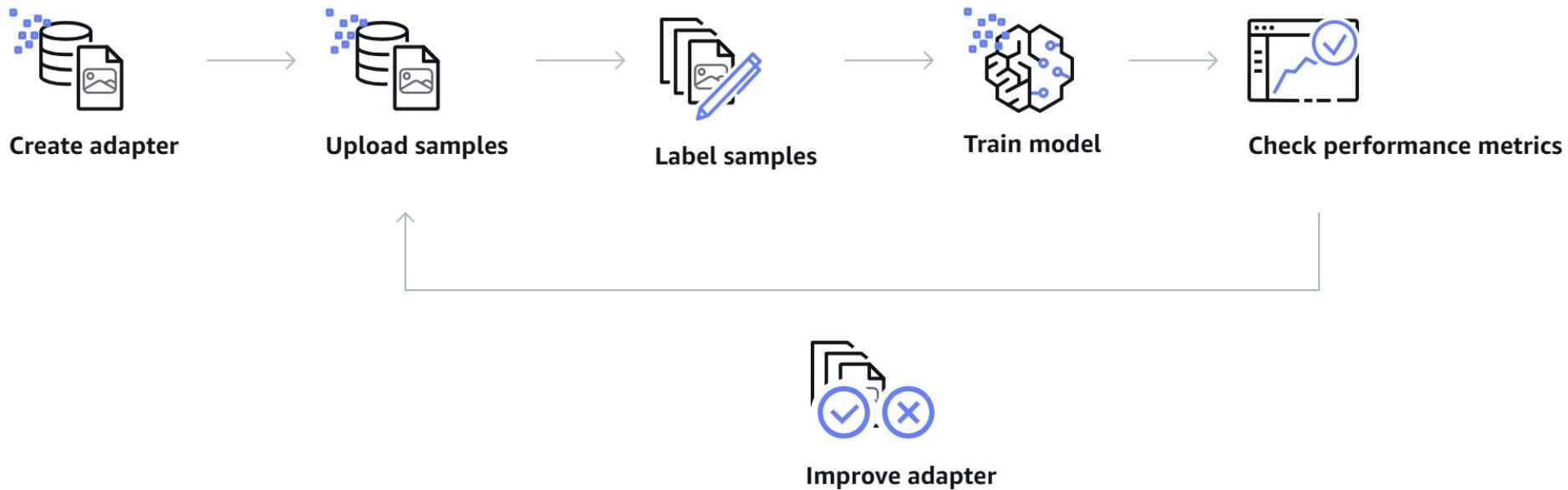
How it works

You provide AWS with:

1. Sample documents (e.g., electricity bills)
2. Ground-truth labels for the fields you care about

Queries adapters

How it works



Adapter Performance

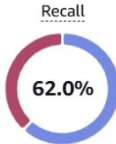
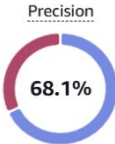
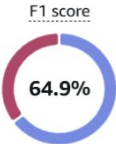
Adapter performance metrics [Info](#)

☐ Switch to baseline metrics

Status
✔ Training Complete

Date created
June 15, 2026, 12:28:58 (UTC+05:30)

Test Documents
30



Performance per query

Performance per document

Performance per page

Performance per query (11) [Info](#)

< 1 2 >

Query	F1 score	Precision	Recall
what is the name	0.380952380952381	0.4	0.363636363636365
what is the bill date	0.8035714285714286	0.8181818181818182	0.7894736842105263
what is the electricity rate per unit	0.8545454545454545	0.8545454545454545	0.8545454545454545
what is the sanction load unit	0.5454545454545454	0.5454545454545454	0.5454545454545454
what is the sanction load	0.5818181818181818	0.5818181818181818	0.5818181818181818
what is the consumer address	0.3478260869565218	0.5	0.2666666666666666
what is the pincode	0.6545454545454545	0.6545454545454545	0.6545454545454545
what is the consumer number or account number	0.8000000000000002	0.8	0.8
what are the arrears	0.9454545454545454	0.9454545454545454	0.9454545454545454
how many units were consumed	0.8181818181818182	0.8181818181818182	0.8181818181818182

AWS Textract + LLM vs Gemini vs AWS Textract

Overall Accuracy=Total Fields Evaluated/ Total Manual Fields Extracted ×100

Tested on 100 documents

	Consumer Number	Consumer Name	Consumer Address	Pincode	Sanction Load	Sanction Load Unit	Total bill amount	bill date	consumed units	arrears	rate per unit	overall accuracy
Match Percentage(aws textract +llama)	90.48	79.17%	61.7	82.61	95.12	95.12	83.33	89.13	87.5	64.71	80	75.49924545
Match Percentage [Gemini (LLMLite + Pr	90.48	95.83%	72.34	39.13	92.68	92.68	79.12	80.41	80.56	76.92	0	64.11620909
Match Percentage [AWS Textract- analyz	90.48	93.75	70.21	35.26	92.68	92.68	81.25	82.65	77.78	69.23	0	71.45181818
manual samples	42	48	47	23	41	41	48	46	40	17	50	

AWS Textract+LLM = 75.49%

Gemini LLM = 64.11%

AWS Textract= 71.45%

Mathematical Deductions

Gemini LLM

- Overall Accuracy: **64.11%**
- Strong on:
 - Arrears
 - Net Bill Amount
 - Sanction Load
- Weak on:

Rate per Unit (0%)

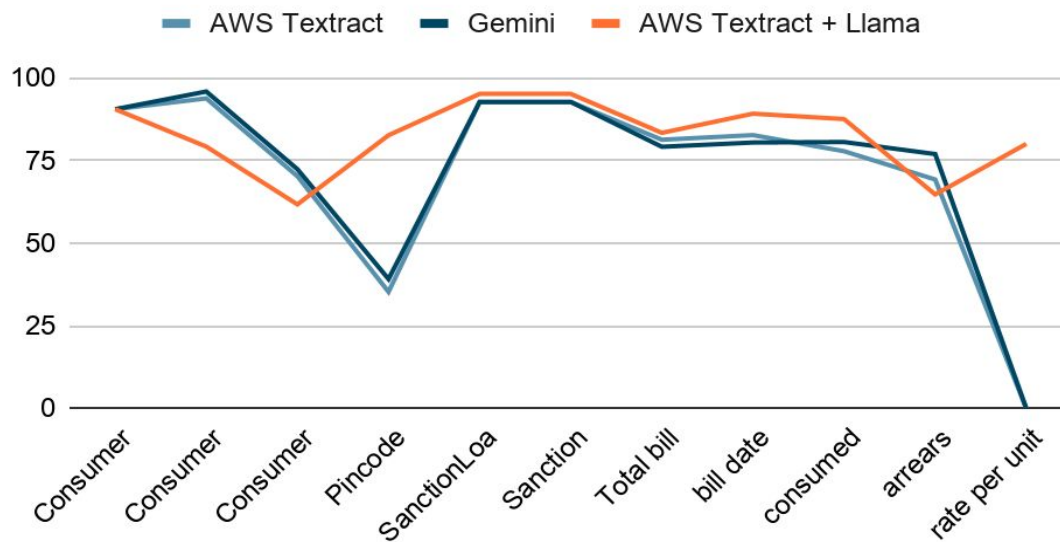
AWS Textract

- Overall Accuracy: **71.23%**
- Similar performance to Gemini.
- Slightly weaker on Consumed Units and Arrears.

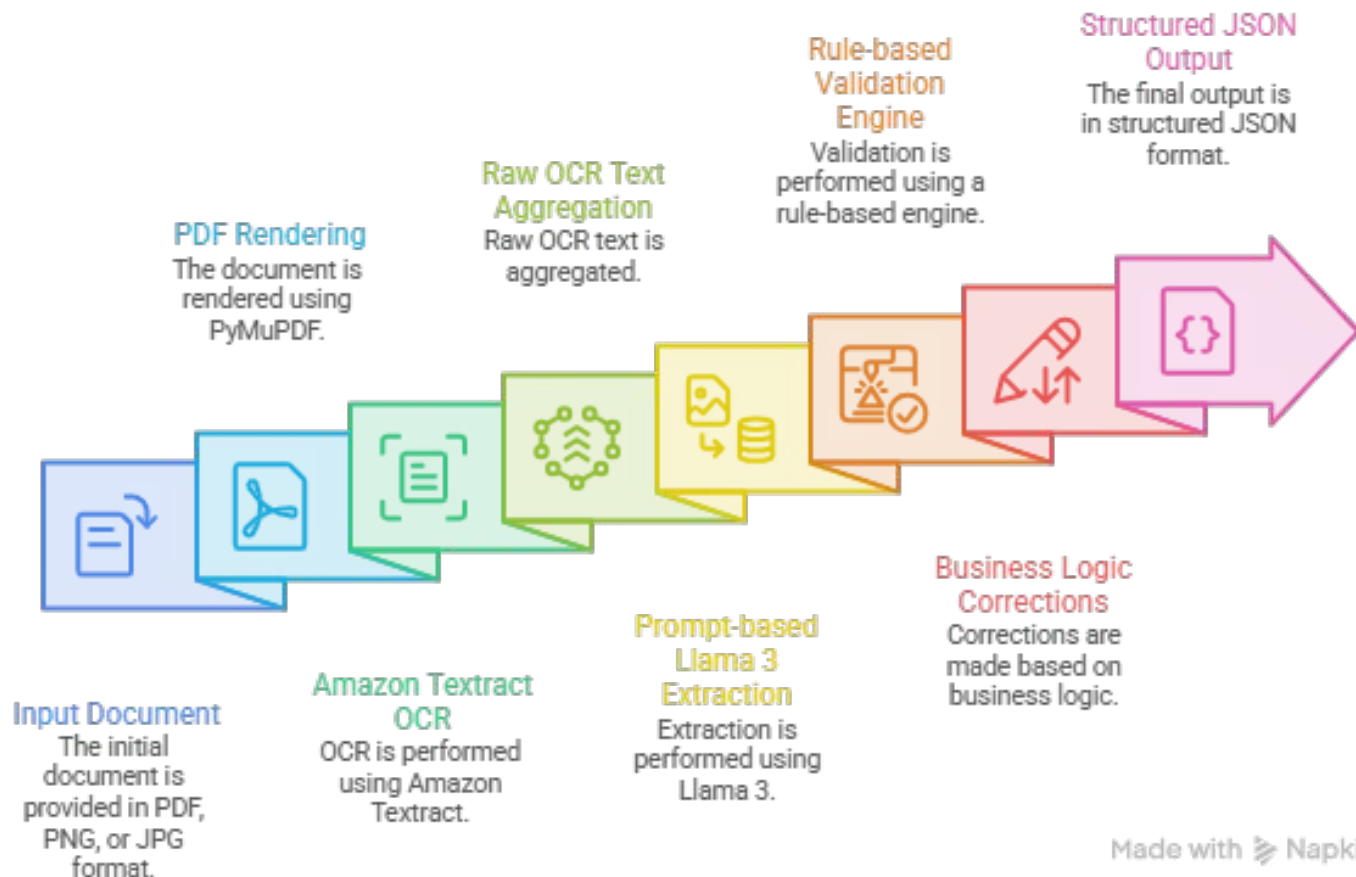
AWS Textract + Llama

- Overall Accuracy: **75.49% (Best Overall)**
- Significant improvements in:
 - Pincode
 - Total Bill Amount
 - Bill Date
 - Consumed Units
 - Rate per Unit
- Weakness:
 - Consumer Name
 - Net Bill Amount

AWS Textract, Gemini and AWS Textract + Llama



Workflow of AWS Textract+ LLM



Technical Highlights

- **Pydantic-Driven JSON Mode:** Rather than relying purely on prompt heuristics, the script dumps a formal Pydantic schema structure directly into the LLM system prompt context alongside the structural API parameter `response_format={"type": "json_object"}`. This guarantees zero schema drift.
- **Hybrid Intelligence & Fallback Calculus:** The pipeline doesn't rely entirely on the LLM's weak mental arithmetic capability for calculating ratios. If `rate_per_unit` is omitted by the model, a native Python Regex routine overrides it via the fallback logic:
- **Confidence Calibration Matrix:** Implements a dual-layer uncertainty profile strategy. It forces the LLM to output custom deterministic floats (0-1) across specific feature elements while dynamically assigning a high baseline confidence factor (0.95) whenever the deterministic string cleaning module handles calculation fallbacks.

DETECT_DOCUMENT APPLICATION

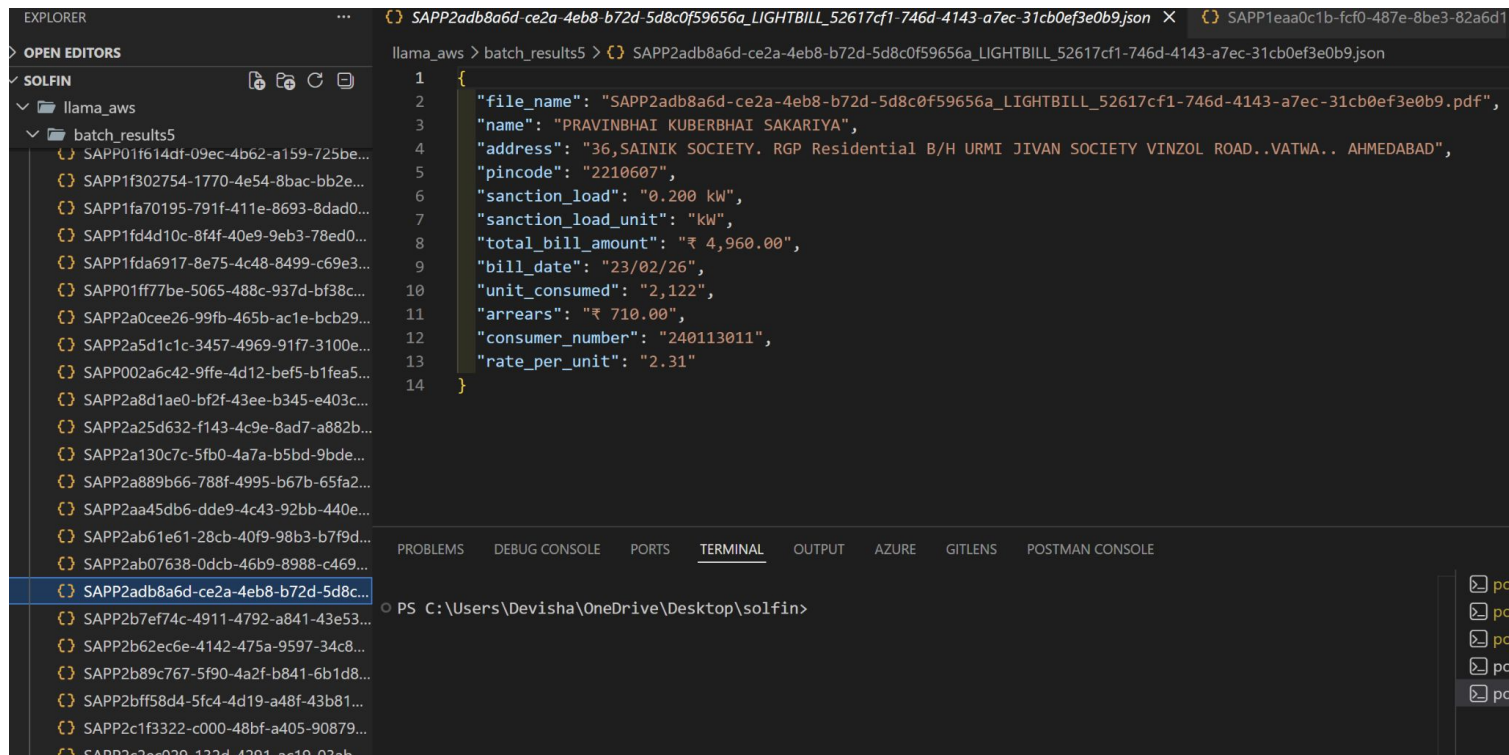
Relying solely on `detect_document_text` for highly structured forms was a weak approach because it strips out the spatial relationships, tables, and form key-value mappings.

However, because your architecture pairs Textract with a state-of-the-art Large Language Model (`llama-3.3-70b-versatile`), the engineering dynamics completely change:

- **Lower Cost & Latency:** `detect_document_text` is significantly cheaper and faster than Textract's advanced feature extraction methods.
- **Prevents Token Waste:** Advanced methods return massive, complex JSON trees containing coordinate pairs, table matrices, and key relationships. Feeding that bloated metadata into an LLM wastes input token limits and adds noise.
- **LLMs Excel at Context Reassembly:** Instead of paying AWS to organize the data into keys and tables, you pass cheap, raw text lines to Llama 3, which is natively exceptional at reading unmapped text and mapping it precisely to a target Pydantic schema.

Week 4

Script for 1000 documents for model testing



The screenshot shows the Visual Studio Code interface. On the left, the Explorer sidebar displays a file tree with the following structure:

- EXPLORED
- OPEN EDITORS
- SOLFIN
 - llama_aws
 - batch_results5
 - SAPP01f614df-09ec-4b62-a159-725be...
 - SAPP1f302754-1770-4e54-8bac-bb2e...
 - SAPP1fa70195-791f-411e-8693-8dad0...
 - SAPP1fd4d10c-8f4f-40e9-9eb3-78ed0...
 - SAPP1fda6917-8e75-4c48-8499-c69e3...
 - SAPP01ff77be-5065-488c-937d-bf38c...
 - SAPP2a0cee26-99fb-465b-ac1e-bcb29...
 - SAPP2a5d1c1c-3457-4969-91f7-3100e...
 - SAPP002a6c42-9ffe-4d12-bef5-b1fea5...
 - SAPP2a8d1ae0-bf2f-43ee-b345-e403c...
 - SAPP2a25d632-f143-4c9e-8ad7-a882b...
 - SAPP2a130c7c-5fb0-4a7a-b5bd-9bde...
 - SAPP2a889b66-788f-4995-b67b-65fa2...
 - SAPP2aa45db6-dde9-4c43-92bb-440e...
 - SAPP2ab61e61-28cb-40f9-98b3-b7f9d...
 - SAPP2ab07638-0dcb-46b9-8988-c469...
 - SAPP2adb8a6d-ce2a-4eb8-b72d-5d8c...
 - SAPP2b7ef74c-4911-4792-a841-43e53...
 - SAPP2b62ec6e-4142-475a-9597-34c8...
 - SAPP2b89c767-5f90-4a2f-b841-6b1d8...
 - SAPP2bff58d4-5fc4-4d19-a48f-43b81...
 - SAPP2c1f3322-c000-48bf-a405-90879...
 - SAPP2c2ec029-132d-4291-ac19-03ab...

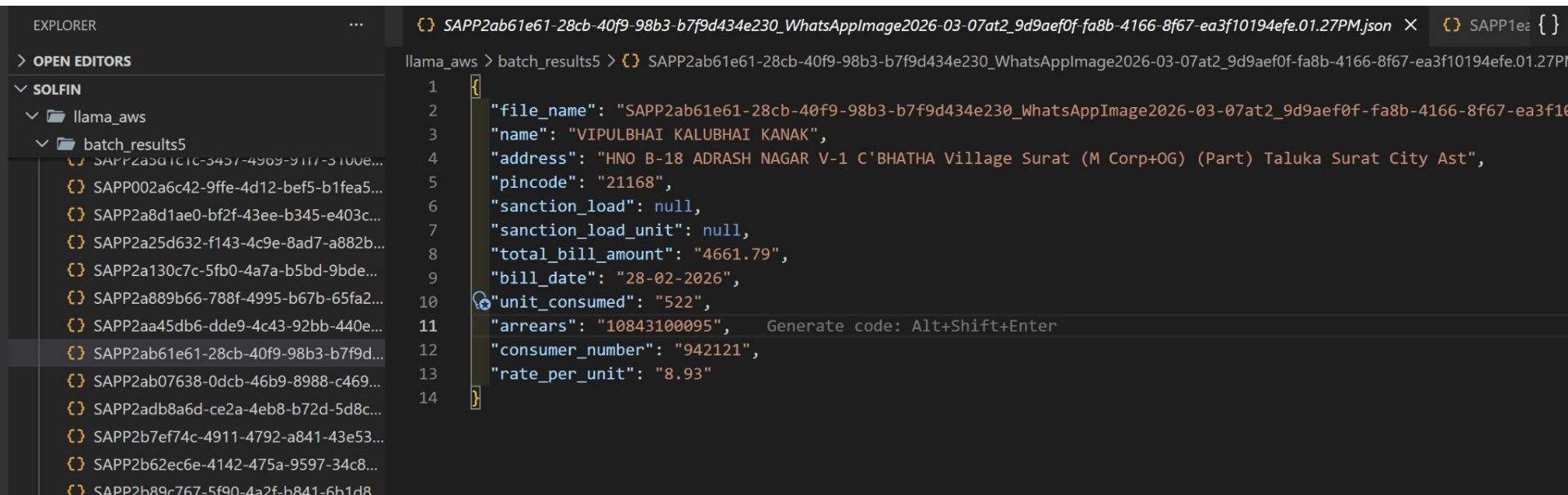
The main editor area shows the content of the selected file, `SAPP2adb8a6d-ce2a-4eb8-b72d-5d8c0f59656a_LIGHTBILL_52617cf1-746d-4143-a7ec-31cb0ef3e0b9.json`. The JSON content is as follows:

```
1 {
2   "file_name": "SAPP2adb8a6d-ce2a-4eb8-b72d-5d8c0f59656a_LIGHTBILL_52617cf1-746d-4143-a7ec-31cb0ef3e0b9.pdf",
3   "name": "PRAVINBHAI KUBERBHAI SAKARIYA",
4   "address": "36,SAINIK SOCIETY. RGP Residential B/H URMI JIVAN SOCIETY VINZOL ROAD..VATWA.. AHMEDABAD",
5   "pincode": "2210607",
6   "sanction_load": "0.200 kW",
7   "sanction_load_unit": "kW",
8   "total_bill_amount": "₹ 4,960.00",
9   "bill_date": "23/02/26",
10  "unit_consumed": "2,122",
11  "arrears": "₹ 710.00",
12  "consumer_number": "240113011",
13  "rate_per_unit": "2.31"
14 }
```

The bottom status bar shows the terminal with the command: `PS C:\Users\Devisha\OneDrive\Desktop\solfin>`

Created a separate model with AWS Textract and regex validation only

The extracted texts are not correctly associated with the data points



```
EXPLORER
...
OPEN EDITORS
SOLFIN
  llama_aws
    batch_results5
      SAPP2a5d1c1c-3457-4969-9117-3100e...
      SAPP002a6c42-9ffe-4d12-bef5-b1fea5...
      SAPP2a8d1ae0-bf2f-43ee-b345-e403c...
      SAPP2a25d632-f143-4c9e-8ad7-a882b...
      SAPP2a130c7c-5fb0-4a7a-b5bd-9bde...
      SAPP2a889b66-788f-4995-b67b-65fa2...
      SAPP2aa45db6-dde9-4c43-92bb-440e...
      SAPP2ab61e61-28cb-40f9-98b3-b7f9d...
      SAPP2ab07638-0dcb-46b9-8988-c469...
      SAPP2adb8a6d-ce2a-4eb8-b72d-5d8c...
      SAPP2b7ef74c-4911-4792-a841-43e53...
      SAPP2b62ec6e-4142-475a-9597-34c8...
      SAPP2b89c767-5f90-4a2f-b841-6b1d8...

SAPP2ab61e61-28cb-40f9-98b3-b7f9d434e230_WhatsAppImage2026-03-07at2_9d9aef0f-fa8b-4166-8f67-ea3f10194efe.01.27PM.json
SAPP1e... {}

llama_aws > batch_results5 > SAPP2ab61e61-28cb-40f9-98b3-b7f9d434e230_WhatsAppImage2026-03-07at2_9d9aef0f-fa8b-4166-8f67-ea3f10194efe.01.27PM.json
1 {
2   "file_name": "SAPP2ab61e61-28cb-40f9-98b3-b7f9d434e230_WhatsAppImage2026-03-07at2_9d9aef0f-fa8b-4166-8f67-ea3f10194efe.01.27PM.json",
3   "name": "VIPULBHAI KALUBHAI KANAK",
4   "address": "HNO B-18 ADRASH NAGAR V-1 C'BHATHA Village Surat (M Corp+OG) (Part) Taluka Surat City Ast",
5   "pincode": "21168",
6   "sanction_load": null,
7   "sanction_load_unit": null,
8   "total_bill_amount": "4661.79",
9   "bill_date": "28-02-2026",
10  "unit_consumed": "522",
11  "arrears": "10843100095",
12  "consumer_number": "942121",
13  "rate_per_unit": "8.93"
14 }
```

Playground

Upload a document and run analysis to test your prompt. Testing as **electricity_bill**.

 Google Gemini / gemini/gemini-2... active ▾

Edit prompt to compare > cURL ▶ Run Current ▶ Run Edited 🔗 Compare

DOCUMENT PREVIEW

 Verification Data ▾

 SAPP1e94cf35-cb48-4475-a3fd-ef40f382310b_Mahavitaran-LTE-BillMAHARU... 1068 KB ✕

< 1 / 4 > 🔍 100% 🔍

Website: www.mahadiscom.in GSTIN of MSEDCCL 27AAECM2933K12B BILL NO.(GGN): 000003200714651 (Opted for Go-Green)		HSN code 27160000	
घाहक क्रमांक: 550011814701 KAMALBAI DEVIDASRAO LOHAR PLOT NO 17 JINDAMNAGAR NANDED Gopalchavdi 431603 मोबा/होम: 88*****24/DLO*****@GMAIL.COM		देयक दिनांक: 26-JAN-26 देयक रकम रु: 16,960.00	
विनिर्माण मुद्रित: 4724 -NANDED (U) MIDC SDN. दर संकेत: 090/LT1 Res 1-Phase फोन नं: 00000013 पी.सी./घर-भारत-क्रम/डि.टी.सी.: 1 / 03-0310-1001 /4724535 मिटर क्रमांक: 08601328142 रिटिंग यु: U1		पुरवठा दिनांक: 02-Jun-2023 मंजूर भार: 3 KW पुरवा देन जमा(रु): 0.00 घातु रिडिंग दिनांक: 23-JAN-26 मागील रिडिंग दिनांक: 22-DEC-25	
		QR code is valid upto: 04-03-2026 Scan this QR Code with BHIM App for UPI Payment	

OUTPUT

Run analysis to see output
Upload a document and click Run

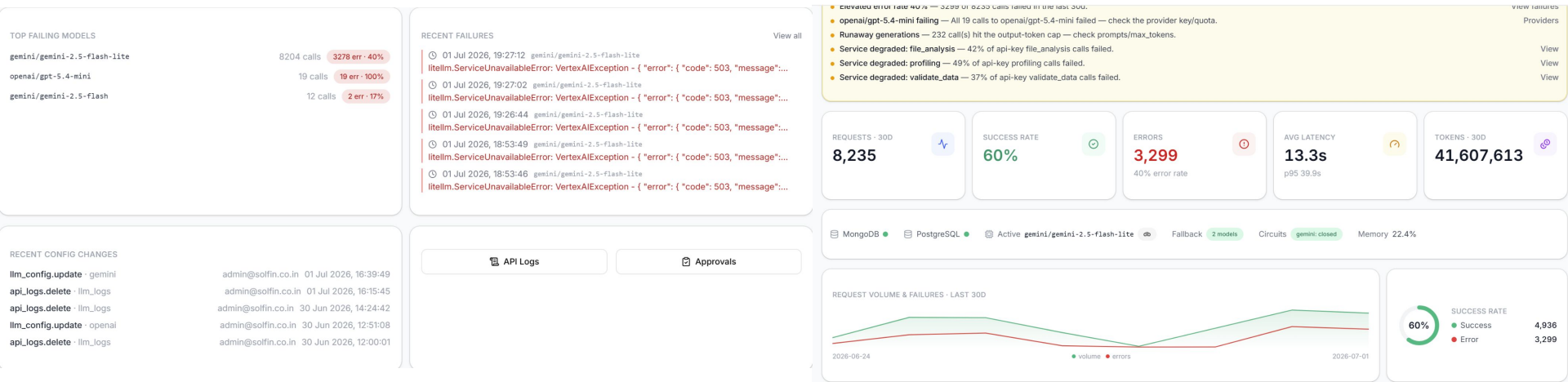
litellm.ServiceUnavailableError: VertexAIException
- { "error": { "code": 503, "message": "This model is currently experiencing high demand. Spikes in demand are usually temporary. Please try again later.", "status": "UNAVAILABLE" } } }

Week 5

The issue is with **Gemini 2.5 Flash Lite**, which handled almost all traffic and generated thousands of failures.

The **GPT-5.4 Mini** endpoint failed on every request, suggesting problems such as:

- Invalid API credentials
- Insufficient quota
- Incorrect endpoint configuration
- Provider outage



Week 6

Data Point	Size Documents + LLM	Gemini		Method	Correct Matches	Total Compared	Overall Accuracy
Consumer Number	95.90% (281/293)	58.36% (171/293)		Analyze Documents	2,738	2,916	89.90%
Name	99.10% (329/332)	99.40% (330/332)		Gemini	2,186	2,916	74.97%
Address	99.04% (308/311)	96.46% (300/311)					
Pincode	98.57% (207/210)	11.90% (25/210)					
Bill Date	98.39% (305/310)	96.45% (299/310)					
Sanction Load	85.87% (243/283)	79.51% (225/283)					
Sanction Load U	86.43% (242/280)	80.36% (225/280)					
Unit Consumed	83.08% (216/260)	78.85% (205/260)					
Total Bill Amount	99.03% (307/310)	95.81% (297/310)					
Rate per Unit	89.13% (205/230)	13.04% (30/230)					
Arrears	97.94% (95/97)	81.44% (79/97)					

Solfin Layout-Agnostic Extractor

Pattern A: Amazon Textract (Analyze Document: Forms + Tables) + Groq Cloud (Llama 3 Semantic Structuring)

Upload Utility Bill Image / PDF File

 SAPP1e2...e16f95.pdf
115.8KB

+

Document successfully structured into explicit key targets!

Overall Extraction Confidence

97.4%

JSON Output Frame

```
{
  "name": "PRABHU SHANKAR SHUKLA"
  "address":
    "OLD MARKET PURAMUFTI ALLAHABAD UP IND"
  "pincode": NULL
  "sanction_load": "4"
  "sanction_load_unit": "kw"
```

Raw OCR

Extracted

... P
[FOI
Prev
Curr

 SAPP01f...16e760.pdf
1.0MB

+

Document successfully structured into explicit key targets!

Overall Extraction Confidence

86.9%

JSON Output Frame

```
{
  "name": "SHRI RAMESH BACHULAL HIRANWAR"
  "address":
    "PILI NADI KAMPTEE ROAD POLE NO HTP/11 NA  
GPUR 17 440017"
  "pincode": "440017"
  "sanction_load": "0.4"
  "sanction_load_unit": "KW"
  "total_bill_amount": "810.00"
  "bill_date": "21-DEC-25"
  "unit_consumed": "97"
```

Raw OCR Text (Textract Analyze)

Extracted Forms, Tables & Lines

--- PAGE 1 ---
[FORMS]
Nov: (Textract Confidence: 0.00)
Oct: (Textract Confidence: 0.00)
Aug: (Textract Confidence: 0.00)
May: (Textract Confidence: 0.00)
Apr: (Textract Confidence: 0.00)
Mar: (Textract Confidence: 0.00)
Dec: (Textract Confidence: 0.00)
2025: (Textract Confidence: 0.00)
2024: (Textract Confidence: 0.00)

Human-In-The-Loop Data Validation

Name (60% Conf.)	Address (60% Conf.)
SHRI RAMESH BACHULAL HIRANW	PILI NADI KAMPTEE ROAD POLE NO HTP/11 NAGPUR 17 440017
Consumer Number (100% Conf.)	Pincode (60% Conf.)
410010929702	440017
Total Bill Amount (100% Conf.)	Sanction Load (98% Conf.)
810.00	0.4
Bill Date (100% Conf.)	Sanction Load Unit (98% Conf.)
21-DEC-25	

Documented these bills as received on real life test cases.

42	aws textract	05026393_ELEC	14405/06354/0	SASHIKANT K	D-1 5 DARSHAN	300009		4642.49	01/06/2026	2
	manual		14405/06354/0	SASHIKANT K	D-1 5 DARSHAN	300009	1 kW	4642.49	01/06/2026	2
		Data Point Field	Total Pairs Eval	Total Matches	Match Percentage					
		bill_date	42	42	100.00%	Total Data Points Evaluated: 495				
		pincode	42	41	97.78%					
		total_bill_amou	42	41	97.78%	Total Exact Matches: 443				
		address	42	40	95.56%					
		consumer_num	42	39	93.33%	Overall Model Accuracy: 89.49%				
		name	42	39	93.33%					
		unit_consumed	42	35	88.89%					
		arrears	42	35	88.89%					
		rate_per_unit	42	34	86.67%					
		sanction_load_	42	32	75.56%					
		sanction_load	42	28	66.67%					

Github Repo link

<https://github.com/devishasolfin/solfin-ocr-aws>

<https://github.com/devishasolfin/solfin-ocr-aws/tree/main>